MODULE 25 · DOMAIN 4 - ASSEMBLING AND DEPLOYING APPLICATIONS

Access control from serving endpoints

Whose identity a deployed agent uses when it reaches an index, a table or an outside API, and how to choose.

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admin@sthithapragna.com

Every door checks a badge. The only question is whose badge the agent is carrying: the deployer's, the user's, or one you issued by hand.

How a deployed agent authenticates

METHOD	RUNS AS	CREDENTIALS	EFFORT
Automatic passthrough	The deployer	Short-lived, auto-rotated	Low: declare resources
On-behalf-of-user	The end user	User's, with restricted scopes	Medium: scopes, runtime set-up
Manual	Credentials you supply	Long-lived, via secrets	High: you rotate them

ON THE EXAM

Automatic authentication passthrough is Databricks' recommended default. Use the others only for a reason.

Choosing a method for each resource

01

Per-user access?

Or per-user audit? Then on-behalf-of-user.

›

02

Supported?

All resources support automatic? Then automatic.

›

03

Otherwise

Manual authentication, with secrets.

ON THE EXAM

The choice is per resource. One agent can use automatic for its index, on-behalf-of-user for a table, and manual for an outside API.

Declaring resources when logging the agent

RESOURCE CLASS	FOR
`DatabricksServingEndpoint`	LLM and embedding endpoints
`DatabricksVectorSearchIndex`	AI Search indexes
`DatabricksFunction`	Unity Catalog function tools
`DatabricksGenieSpace`	Genie Agents
`DatabricksSQLWarehouse`, `DatabricksTable`	Warehouses and tables
`DatabricksUCConnection`, `DatabricksLakebase`	External connections and Lakebase

ON THE EXAM

Resources go in the resources list of log_model. Databricks then issues credentials scoped to exactly those resources.

Declare everything downstream too

- A **Genie space** needs its **tables, SQL warehouse and functions** declared as well.
- A **managed MCP server** needs every index or function behind it declared.
- databricks_mcp can list the resources a managed MCP server needs.
- Undeclared resources fail only once deployed, not in the notebook.

> ON THE EXAM
>
> Works in the notebook, permission error once deployed: a resource was not declared. The notebook used your own identity.

Automatic passthrough, or on-behalf-of-user?

AUTOMATIC PASSTHROUGH

Everyone sees the same

- Deployer's permissions for every request
- Short-lived credentials, managed for you
- Right when all users may see the same data
- Recommended default

ON-BEHALF-OF-USER

Each user sees their own

- Each request uses the caller's permissions
- Row filters and masks apply per user
- Audit shows who really asked
- Scopes declared, credentials set up per request

THE TIEBREAKER

If different users must see different data, or audit must name the real user, use on-behalf-of-user. Otherwise, automatic.

What it takes

- Declare the **API scopes** the agent needs when logging it.
- Create the user's client **inside predict**, with ModelServingUserCredentials.
- The user's identity only exists during a request, not when the model loads.
- Scopes restrict what the agent can do, even with the user's identity.
- Currently in **Public Preview**.

> ON THE EXAM
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> Set up user credentials per request, in predict. At load time there is no user yet.

When you must supply credentials

- For resources automatic passthrough does not support, such as an outside API.
- Keep the credential in **Databricks secrets**, exposed as an **environment variable**.
- Prefer a **service principal with OAuth** over a personal access token.
- Never paste a key into code or a notebook. It leaks into version control and logs.
- Long-lived credentials need a rotation plan.

> ON THE EXAM
>
> Manual does not mean careless. The secret lives in secrets, and the code only reads an environment variable.

Harbourline's claims agent, three badges

01

Index and LLM

Automatic passthrough: everyone may use them.

02

Claims table

On-behalf-of-user: adjusters see their region.

03

Fraud-score API

Manual: key in secrets, env variable.

04

Genie

Space, tables and warehouse all declared.

05

Callers

CAN QUERY on the endpoint, per team.

ON THE EXAM

One agent, three methods, each chosen per resource using Databricks' flowchart.

> An agent works perfectly in the developer's notebook. After deployment to Model Serving, every request fails with a permission error on its AI Search index.
>
> The agent was logged without a resources list.

What should the engineer do?

A Give every end user direct access to the index itself

B Put the developer's personal access token in agent.py

C Increase the endpoint's workload size, then redeploy

D Declare the index in resources when logging the agent

◆ PAUSE AND DECIDE

What should the engineer do?

A Access for every user

Automatic passthrough uses the deployer, not the users.

B Token in agent.py

A long-lived secret in code. Never.

C Bigger workload

Capacity has nothing to do with permission.

D Declare it in resources

Databricks then issues scoped, short-lived credentials.

THE SIGNAL

Works in the notebook, fails when deployed, no resources list. The notebook used the developer's own identity.

> An HR agent answers questions using an employee records table. Each employee may see only their own record, enforced by a Unity Catalog row filter.
>
> Audit logs must show which employee asked each question.

Which authentication method fits the records table?

A Automatic passthrough, with the deployer's permissions

B On-behalf-of-user authentication, with declared scopes

C Manual authentication, with one shared service key

D A system prompt telling the agent to filter by the user

◆ PAUSE AND DECIDE

Which authentication method fits the records table?

A Automatic passthrough

The deployer's view for everyone, bypassing the per-user filter.

B On-behalf-of-user

Row filter applies per caller, and audit names the real user.

C Manual, shared key

One identity for all: same problem as A, plus rotation.

D System prompt filter

Not a security control.

THE SIGNAL

Only their own record, and audit by employee. Both are the flowchart's first question: per-user access or per-user audit.

> An agent must call an outside weather-risk service that requires an API key.
>
> Automatic authentication passthrough does not support this service.

How should the agent get the key?

A Paste the API key directly into the agent's source code

B Declare the service as a DatabricksServingEndpoint resource

C Store it in secrets and expose it as an environment variable

D Use on-behalf-of-user so each user's own key is sent instead

◆ PAUSE AND DECIDE

How should the agent get the key?

A Key in source code

Leaks into version control and logs.

B Declare as serving endpoint

It is an outside service, not a Databricks endpoint.

C Secret as environment variable

Manual authentication done properly.

D On-behalf-of-user

Passes Databricks identity. Users hold no weather-service key.

THE SIGNAL

Automatic passthrough does not support it, and no per-user need. The flowchart ends at manual, with a secret.

> A deployed agent calls a Genie space. It was logged with DatabricksGenieSpace in its resources, and nothing else.
>
> Genie requests fail when trying to run the generated SQL.

What is the most likely fix?

A. Also declare the Genie space's tables and SQL warehouse

B. Switch the agent to manual auth with a personal token

C. Grant the end users CAN MANAGE on the Genie space

D. Recreate the Genie space inside the agent's own schema

◆ PAUSE AND DECIDE

What is the most likely fix?

A Declare tables and warehouse
Downstream resources must be declared too.

B Manual, personal token
Swaps a managed credential for a long-lived one.

C CAN MANAGE for users
Automatic passthrough does not use users' permissions.

D Recreate the space
Moving it does not declare what it depends on.

THE SIGNAL

The space is declared, but the SQL fails. Genie runs SQL on tables and a warehouse, and those were never declared.

What to carry forward

- **Automatic passthrough**: deployer's permissions, declared resources, recommended default.
- **On-behalf-of-user**: per-user access or audit. Scopes, credentials set up in predict.
- **Manual**: unsupported resources. Secrets as environment variables, never in code.
- Declare **downstream** resources too: Genie's tables and warehouse, MCP's indexes.
- Works in the notebook, fails deployed: a resource was not declared.

> ON THE EXAM
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> Next: module 26 gives agents memory with Lakebase, and tools through MCP.